

ECE 332 — Homework 3 Walkthrough

Cheat sheet snippet → formula template → givens → plug → answer.Problem 1 — 200 MHz LHC, Air → Dielectric ($\epsilon_r = 4$)

20 pts

Setup: 200 MHz LHC plane wave, $|E| = 5$ V/m, normal incidence onto $\epsilon_r = 4$ at $z \geq 0$. Positive max at $z = 0, t = 0$.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

Med. 1 ($z < 0$) incident side; Med. 2 ($z \geq 0$) transmitted.			
	Normal $\perp$ ($\theta_i = 0$)	pol	pol
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{\parallel} ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
$R+T$ total	$R = 1 - T$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$

Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; nonmag. $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$.

Intrinsic impedance η_i plug into table above

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_r, i}} \quad (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction $\eta_c \sigma / (\omega \epsilon') \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega \epsilon'} \right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\epsilon_r}$. The j term is for $|\eta_c|, \theta_{\eta}$. See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

 $\tilde{\mathbf{E}}$ PHASOR TEMPLATES (Air: $\epsilon_r = \mu_r = 1$)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega \sqrt{\epsilon_r, i}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_r, 2 / \epsilon_r, 1}$$

Direction table $u \in \{x, y, z\} = \text{axis perp. to boundary}$

Inc dir	Inc	Refl	Trans
$+\hat{u}$	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
$-\hat{u}$	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u} \text{ (V/m)} \quad (\text{incident wave, Med 1})$$

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u} \text{ (V/m)} \quad (\text{reflected wave, Med 1})$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u} \text{ (V/m)} \quad (\text{transmitted wave, Med 2})$$

(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

(a) Incident $\mathbf{E}$ phasor. Template: $\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{-jk_1 u}$.

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j(4\pi/3)z} \text{ V/m}$$

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

Givens:

$$f = 200 \text{ MHz}, \omega = 2\pi f = 4\pi \times 10^8 \text{ rad/s}$$

$$u = z; \mu_r = \epsilon_r = 1 \text{ (Air)}$$

$$\hat{e} = \hat{x} + j\hat{y} \text{ (LHC)}$$

$$k_1 = \omega \sqrt{\epsilon_r} / c = 4\pi/3 \text{ rad/m}$$

$$c = 3 \times 10^8 \text{ m/s}; a_0 = 5 \text{ V/m}$$

(b) Reflection & transmission coefficients.

$$\eta_1(\text{Air}) = \sqrt{\frac{\mu_r}{\varepsilon_{r1}}} \eta_0 = \sqrt{1/1}(120\pi) = 120\pi \Omega$$

$$\eta_2(\text{med 2}) = \sqrt{\frac{\mu_r}{\varepsilon_{r2}}} \eta_0 = \sqrt{1/4}(120\pi) = 60\pi \Omega$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \boxed{-\frac{1}{3}} \text{ (unitless)}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}} \text{ (unitless)}$$

(c) **Reflected, transmitted, total field.**

Templates: $\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{+jk_1 u}$, $\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{-jk_2 u}$.

$$\tilde{\mathbf{E}}^r = -\frac{1}{3}(5)(\hat{x} + j\hat{y}) e^{+j(4\pi/3)z}$$

$$\boxed{\tilde{\mathbf{E}}^r = -\frac{5}{3}(\hat{x} + j\hat{y}) e^{j4\pi z/3} \text{ V/m}}$$

$$\tilde{\mathbf{E}}^t = \frac{2}{3}(5)(\hat{x} + j\hat{y}) e^{-j(8\pi/3)z}$$

$$\boxed{\tilde{\mathbf{E}}^t = \frac{10}{3}(\hat{x} + j\hat{y}) e^{-j8\pi z/3} \text{ V/m}}$$

$$\tilde{\mathbf{E}}_T = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$$

$$\boxed{\tilde{\mathbf{E}}_T = 5(\hat{x} + j\hat{y}) \left[e^{-j4\pi z/3} - \frac{1}{3} e^{j4\pi z/3} \right] \text{ V/m}}$$

(d) **% reflected and transmitted power.**

$$\%P_r = 100|\Gamma|^2 = 100\left|-\frac{1}{3}\right|^2 = \boxed{11.1\%}$$

$$\%P_t = 100|\tau|^2 \frac{\eta_1}{\eta_2} = 100\left|\frac{2}{3}\right|^2 \frac{120\pi}{60\pi} = \boxed{88.9\%}$$

Check: $11.1\% + 88.9\% = 100\%$. *Energy conserved* ✓

Givens:

$$\eta_0 = 120\pi \Omega \approx 377 \Omega$$

Givens:

$$k_2 = \omega \sqrt{\varepsilon_{r2}} / c = 8\pi/3$$

$$\varepsilon_{r2} = 4; c = 3 \times 10^8$$

$$\omega = 400\pi \times 10^6$$

Problem 2 — Same Setup, Med 2 = Poor Conductor ($\epsilon_r=2.25$, $\sigma=10^{-4}$) 20 pts**Setup:** Replace medium 2 with poor conductor: $\epsilon_r = 2.25$, $\mu_r = 1$, $\sigma = 10^{-4}$ S/m. Med 1 unchanged.**TABLE 8-2 — Γ , τ , R , T (unitless) SUMMARY**

<i>Med. 1 ($z < 0$) incident side; Med. 2 ($z \geq 0$) transmitted.</i>			
	Normal $\perp$ pol ($\theta_i = 0$)		$\parallel$ pol
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{\parallel} ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
$R + T$ total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$

Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; nonmag. $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$.

Intrinsic impedance η_i plug into table above

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \quad (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction $\eta_c \quad \sigma/(\omega\epsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon} \right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\epsilon_r}$. The j term is for $|\eta_c|, \theta_{\eta}$. See LOSSY MEDIA — 5 CASES for all regimes.

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

Type	$\sigma/\omega\epsilon$	η_c, α, β
Lossless ($\sigma=0$)	0	$\eta = \eta_0/\sqrt{\epsilon_r}$ exact $\alpha=0, \beta = \omega\sqrt{\epsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon_r} \right)$ $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	$10^{-2} - 10^2$	exact: $(\eta/\sqrt{X})e^{j\theta_{\eta}}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$

For Γ/τ : drop j correction; use $\eta_0/\sqrt{\epsilon_r}$. Magnetic: keep μ_r .

Low-loss quick params $\sigma/(\omega\epsilon) \ll 1, \mu_r = 1$

$$\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}} \text{ (Np/m)}, \beta \approx \frac{\omega\sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}$$

Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X-1}, \quad \beta = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X+1}$$

$$\theta_{\eta} = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'}$$

 $\tilde{\mathbf{E}}$ PHASOR TEMPLATES (Air: $\epsilon_r = \mu_r = 1$)

Wavenumber per medium nonmag.

$$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table $u \in \{x, y, z\} = \text{axis perp. to boundary}$

Inc dir	Inc	Refl	Trans
$+\hat{u}$	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
$-\hat{u}$	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u} \text{ (V/m)} \quad (\text{incident wave, Med 1})$$

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u} \text{ (V/m)} \quad (\text{reflected wave, Med 1})$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u} \text{ (V/m)} \quad (\text{transmitted wave, Med 2})$$

(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization plug into templates above

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected always

$$\%P_r = 100 |\Gamma|^2$$

% transmitted η -ratio matters!

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

(a) Incident $\mathbf{E}$ phasor. Med 1 didn't change, so same form as P1.

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

(b) Classify medium 2, then Γ, τ .

Loss tangent:

$$\frac{\sigma}{\omega\epsilon'} = \frac{10^{-4}}{(4\pi \times 10^8)(2.25)(8.85 \times 10^{-12})} = 3.99 \times 10^{-3}$$

Since $\sigma/\omega\epsilon' \ll 10^{-2}$, **low-loss**; use $\eta_0/\sqrt{\epsilon_r}$.

$$\eta_1 = 120\pi \Omega, \quad \eta_2 = \frac{120\pi}{\sqrt{2.25}} = 80\pi \Omega$$

$$\Gamma = \frac{80\pi - 120\pi}{80\pi + 120\pi} = \boxed{-\frac{1}{5}} \text{ (unitless)}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{5} = \boxed{\frac{4}{5}} \text{ (unitless)}$$

(c) Reflected, transmitted (lossy), total field.

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e}^{+jk_1 z} = -\frac{1}{5}(5)(\hat{x} + j\hat{y})e^{+j4\pi z/3}$$

$$\boxed{\tilde{\mathbf{E}}^r = -(\hat{x} + j\hat{y})e^{j4\pi z/3} \text{ V/m}}$$

$\tilde{\mathbf{E}}^t$ replaces $e^{-jk_2 z}$ with $e^{-\alpha_2 z}e^{-j\beta_2 z}$ (lossy):

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e}^{-\alpha_2 z} e^{-j\beta_2 z} = \frac{4}{5}(5)(\hat{x} + j\hat{y})e^{-0.004\pi z} e^{-j2\pi z}$$

$$\boxed{\tilde{\mathbf{E}}^t = 4(\hat{x} + j\hat{y})e^{-0.004\pi z} e^{-j2\pi z} \text{ V/m}}$$

$$\boxed{\tilde{\mathbf{E}}_T = 5(\hat{x} + j\hat{y})\left[e^{-j4\pi z/3} - \frac{1}{5}e^{j4\pi z/3}\right] \text{ V/m}}$$

(d) % reflected and transmitted power.

$$\%P_r = 100|\Gamma|^2 = 100\left|-\frac{1}{5}\right|^2 = \boxed{4\%}$$

$$\%P_t = 100|\tau|^2 \frac{\eta_1}{\eta_2} = 100\left|\frac{4}{5}\right|^2 \frac{120\pi}{80\pi} = \boxed{96\%}$$

Check: 4% + 96% = 100%. Energy conserved ✓

Givens (med 2):

$$\epsilon_{r2} = 2.25, \quad \mu_{r2} = 1$$

$$\sigma_2 = 10^{-4} \text{ S/m}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$\omega = 400\pi \times 10^6 \text{ rad/s}$$

$$\eta_0 = 120\pi \Omega$$

Low-loss row (Table 5-cases):

$$\alpha_2 \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}} = \frac{10^{-4}(120\pi)}{2\sqrt{2.25}} = 0.004\pi$$

$$= 1.26 \times 10^{-2} \text{ Np/m}$$

$$\beta_2 \approx \frac{\omega\sqrt{\epsilon_r}}{c} = \frac{(400\pi \times 10^6)(1.5)}{3 \times 10^8} = 2\pi \text{ rad/m}$$

Problem 3 — Parallel-Pol, Two Layers ($\epsilon_r=6.25, 2.25$)**10 pts****Setup:** $\theta_i = 30^\circ$ from air into layer 1 ($\epsilon_r = 6.25, 5 \text{ cm}$), layer 2 ($\epsilon_r = 2.25, 5 \text{ cm}$), exits to air.**SNELL'S LAW — OBLIQUE INCIDENCE****Single boundary** *angles from normal*

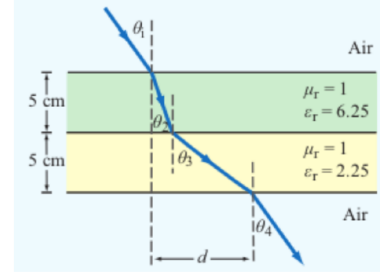
$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4*

$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

*Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).***LATERAL DISPLACEMENT****Beam offset through N slabs** *thickness t_i , refracted angle inside slab i* **Slab-indexed** θ_i = angle inside slab i ; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

*The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).***(a) Chained Snell's law:** $\sin \theta_{i+1} = \sin \theta_i \sqrt{\epsilon_{r,i}/\epsilon_{r,i+1}}$

$$\sin \theta_2 = \sin 30^\circ \sqrt{1/6.25} = 0.5(0.4) = 0.2$$

$$\theta_2 = \sin^{-1}(0.2) = 11.54^\circ$$

$$\sin \theta_3 = \sin \theta_2 \sqrt{6.25/2.25} = 0.2(5/3) = 1/3$$

$$\theta_3 = \sin^{-1}(1/3) = 19.47^\circ$$

$$\sin \theta_4 = \sin \theta_3 \sqrt{2.25/1} = (1/3)(1.5) = 0.5$$

$$\theta_4 = 30.00^\circ = \theta_i$$

(b) Lateral distance $d = \sum_i t_i \tan \theta_{\text{inside slab } i}$

$$d = t_2 \tan \theta_2 + t_3 \tan \theta_3$$

$$= 0.05 \tan(11.54^\circ) + 0.05 \tan(19.47^\circ)$$

$$= 0.05(0.2041) + 0.05(0.3534)$$

$$d = 0.0279 \text{ m} = 2.79 \text{ cm}$$

Givens:

$$\epsilon_{r1} = 1 \text{ (air, } \theta_1 = 30^\circ)$$

$$\epsilon_{r2} = 6.25 \text{ (layer 1)}$$

$$\epsilon_{r3} = 2.25 \text{ (layer 2)}$$

$$\epsilon_{r4} = 1 \text{ (air exit)}$$

Givens:

$$t_2 = t_3 = 0.05 \text{ m (slab thickness)}$$

$$\theta_2 = 11.54^\circ \text{ (inside layer 1)}$$

$$\theta_3 = 19.47^\circ \text{ (inside layer 2)}$$

Problem 4 — TE Waveguide, Find Mode, ε_r , f_c , H_y

20 pts

Setup: TE wave, $a = 5$ cm, $b = 3$ cm, $E_x = -36 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z)$ V/m.

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0$, $H_z \neq 0$

TM mode $H_z = 0$, $E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) read m, n from arguments

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE_{mn} vs TM_{mn} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” $\rightarrow$ TE).

2. If $m = 0$ or $n = 0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).

3. Source field $H_z(x, y)$ given $\Rightarrow$ TE. Source field $E_z(x, y)$ given $\Rightarrow$ TM.

TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. That's why TE_{10} exists but TM_{10} doesn't.

 ε_r FROM DISPERSION

Given ω, β , mode (m, n) *result is unitless*

$$\varepsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] \text{ (unitless)}$$

Inputs: ω (rad/s), β (rad/m), a, b (m), $c = 3 \times 10^8$ (m/s), m, n (unitless).

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω .

TE mode generator: $\tilde{H}_z$

$$\begin{aligned}\tilde{H}_z &= H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \\ \tilde{E}_x &= \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \\ \tilde{E}_y &= -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \\ \tilde{E}_z &= 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TE}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TE}} \\ Z_{\text{TE}} &= \eta / \sqrt{1 - (f_c/f)^2}\end{aligned}$$

TM mode generator: $\tilde{E}_z$

$$\begin{aligned}\tilde{E}_z &= E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \\ \tilde{E}_x &= -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \\ \tilde{E}_y &= -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \\ \tilde{H}_z &= 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TM}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TM}} \\ Z_{\text{TM}} &= \eta \sqrt{1 - (f_c/f)^2}\end{aligned}$$

TEM plane wave for reference

$$\begin{aligned}\tilde{E}_x &= E_0 e^{-j\beta z}, \quad \tilde{E}_y = \tilde{E}_z = 0 \\ \tilde{H}_y &= \tilde{E}_x/\eta, \quad \tilde{H}_x = \tilde{H}_z = 0 \\ \eta &= \sqrt{\mu/\epsilon} (\Omega), \quad f_c \text{ N/A}, \quad k = \omega\sqrt{\mu\epsilon}\end{aligned}$$

Properties common to TE/TM

$$\begin{aligned}f_c &= \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)} \\ \beta &= k \sqrt{1 - (f_c/f)^2} \text{ (rad/m)} \\ u_p &= \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)} \\ u_{p0} &= 1/\sqrt{\mu\epsilon} \text{ (m/s)}\end{aligned}$$

New: Table 8-3 now contains $Z_{\text{TE}}/Z_{\text{TM}}$, full TEM column, and Properties Common (f_c , β , u_p , u_{p0})

— *one stop for all waveguide field formulas.*

(a) **Mode number.** Match E_x to Table 8-3 TE-row form $\cos(m\pi x/a) \sin(n\pi y/b)$.

$$\frac{m\pi}{a} = 40\pi \Rightarrow m = 40(0.05) = 2$$

$$\frac{n\pi}{b} = 100\pi \Rightarrow n = 100(0.03) = 3$$

$$\boxed{\text{Mode: TE}_{23}}$$

Givens:

$$a = 0.05 \text{ m}, b = 0.03 \text{ m}$$

TE wave (given), so $E_z = 0$

(b) ε_r from ε_r **FROM DISPERSION** box (Example 8-8).

$$\begin{aligned} \varepsilon_r &= \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right] \\ &= \left(\frac{3 \times 10^8}{2.4\pi \times 10^{10}} \right)^2 \left[(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2 \right] \end{aligned}$$

$$\boxed{\varepsilon_r = 2.25}$$

Givens:

$$\beta = 52.9\pi \text{ rad/m}$$

$$\omega = 2.4\pi \times 10^{10} \text{ rad/s}$$

$$c = 3 \times 10^8 \text{ m/s}$$

(c) **Cutoff frequency for TE₂₃ from Table 8-3 Properties Common.**

$$\begin{aligned} u_{p0} &= \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m/s} \\ f_{23} &= \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2} \\ &= 10^8 \sqrt{(2/0.05)^2 + (3/0.03)^2} = 10^8 \sqrt{11600} \end{aligned}$$

$$\boxed{f_c = 10.77 \text{ GHz}}$$

Givens:

$$m = 2, n = 3$$

$$a = 0.05, b = 0.03 \text{ m}$$

$$\varepsilon_r = 2.25 \text{ (from b)}$$

(d) H_y from Table 8-3 **TE** block: $\tilde{H}_y = \tilde{E}_x / Z_{\text{TE}}$.

$$\begin{aligned} \eta &= \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{2.25}} = 80\pi \Omega \\ Z_{\text{TE}} &= \frac{\eta}{\sqrt{1 - (f_c/f)^2}} = \frac{80\pi}{\sqrt{1 - (10.77/12)^2}} = 569.9 \Omega \\ H_y &= \frac{E_x}{Z_{\text{TE}}} = \frac{-36}{569.9} \cos(40\pi x) \sin(100\pi y) \sin(\omega t - \beta z) \end{aligned}$$

Givens:

$$f = \omega/(2\pi) = 12 \text{ GHz}$$

$$f_c = 10.77 \text{ GHz}$$

$$\eta_0 = 120\pi \approx 377 \Omega$$

$$\varepsilon_r = 2.25$$

$$\boxed{H_y = -0.0632 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z) \text{ A/m}}$$

Problem 5 — Hollow Guide Multi-Mode Travel Times

8 pts

Setup: Hollow guide ($\epsilon_r = 1$, $u_{p0} = c$), $a = 3$ cm, $b = 2$ cm. Carrier 9.5 GHz, length $L = 100$ m. Find t for each excited mode.

CUTOFF FREQUENCY

Common cutoffs ($a > b$) f_c formula in Table 8-3

Mode	f_{mn}
u_{p0}	$c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$)
TE ₁₀	$f_{10} = u_{p0}/(2a)$ (dominant)
TE ₀₁	$f_{01} = u_{p0}/(2b)$
TE ₂₀	$f_{20} = u_{p0}/a = 2f_{10}$
TE ₁₁ , TM ₁₁	$f_{11} = \frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$. In Z_{TE}/Z_{TM} and u_g , $f_c = f_{mn}$ of the excited mode.

GROUP VELOCITY & TRAVEL TIME

Practical $\eta_0 = 120\pi$; nonmag. media

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} (\Omega) \quad (\Leftrightarrow \sqrt{\mu/\epsilon} \text{ in Tab. 8-3})$$

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$

$$u_p u_g = u_{p0}^2 \text{ (m}^2/\text{s}^2\text{)}$$

$t = L/u_g$ (s) travel time through guide of length L
 Higher modes have lower $u_g \Rightarrow$ arrive later. (u_p , β , f_c , $Z_{TE/TM}$ in Tab. 8-3.)

Cutoffs (compute lowest modes, stop when $f_{mn} > 9.5$ GHz):

$$f_{10} = \frac{c}{2a} = \frac{3 \times 10^8}{0.06} = 5 \text{ GHz} \quad (< 9.5, \text{ excited})$$

$$f_{01} = \frac{c}{2b} = \frac{3 \times 10^8}{0.04} = 7.5 \text{ GHz} \quad (< 9.5, \text{ excited})$$

$$f_{20} = \frac{c}{a} = 10 \text{ GHz} \quad (> 9.5, \text{ not excited})$$

$$f_{11} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = 9.014 \text{ GHz} \quad (< 9.5, \text{ excited})$$

Givens:

$f = 9.5$ GHz (carrier)

$a = 0.03$, $b = 0.02$ m

$u_{p0} = c$ (hollow)

$L = 100$ m

Travel times: $u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$, $t = L/u_g$.

Mode	f_c (GHz)	u_g (m/s)	$t = L/u_g$
TE ₁₀	5.000	$3 \times 10^8 \sqrt{1 - (5/9.5)^2} = 2.551 \times 10^8$	0.392 μ s
TE ₀₁	7.500	$3 \times 10^8 \sqrt{1 - (7.5/9.5)^2} = 1.841 \times 10^8$	543.1 ns
TE ₂₀	10.000	N/A (not excited)	N/A
TE ₁₁ , TM ₁₁	9.014	$3 \times 10^8 \sqrt{1 - (9.014/9.5)^2} = 9.510 \times 10^7$	1.05 μ s

Higher modes arrive later: as $f_{mn} \rightarrow f$, $u_g \rightarrow 0$, so closer-to-cutoff modes have longer travel time.